MILITARY SPECIFICATION

MULTIMETER, BATTERY

TESTING: 0- TO 35-AMP,

0- TO 10-VOLT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a watch-type, self-contained multimeter.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	- Boxes, Folding, Paperboard.
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Boxes, Shipping, Fiberboard.
PPP-B-640	- Boxes, Fiberboard, Corrugated, Triple-Wall.
PPP-B-676	- Boxes, Setup.

Military

MIL-P-116	- Preservation-Packaging, Methods of.

FSC 6625

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

STANDARDS

Military

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity, or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Tariff Order Section, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Description. The multimeter shall be a single-dial, dual-scale watch-type voltammeter capable of measuring voltages from 0 to 10 volts and amperages from 0 to 35 amps (see Figure 1 and 6.3).

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

3.2 Material. Material shall be as specified herein. Materials not specified shall be selected by the supplier, and shall be subject to all provisions of this specification.

3.3 Case. The case shall be of corrosion-resistant metal and shall be moisture-proof and dust-proof. It shall be approximately 2-1/4 inches in diameter and 5/8 inch thick. Terminals shall be mounted on the exterior of the case.

3.4 Electric circuits. The voltmeter and ammeter circuits shall produce indicator readings within industry standards of accuracy. The voltmeter circuit shall be a low-resistance type designed to accurately reflect a battery's voltage capacity under load.

3.5 Dial and pointer. The dial shall be clearly marked with scale divisions of not more than 1/5 of a volt for the 0 to 10 voltage scale and not more than one ampere for the 0 to 35 amperage scale. The pointer shall indicate the scale reading and shall be in sharp contrast with the dial background.

3.6 Window. A transparent window shall cover the dial and pointer.

3.7 Terminals and leads. Separately marked voltage and amperage terminals and a common ground terminal shall be attached to the exterior of the case. An insulated, flexible lead not less than 6 inches long shall be attached to the ground terminal. The ground terminal may be electrically common with the metal case.

3.8 Identification marking. The multimeter shall be identified in accordance with MIL-STD-130.

3.9 Treatment and painting. Treatment and painting shall be in accordance with the suppliers standard practice.

3.10 Workmanship. The dial shall be mounted and secured to prevent movement or rattling in the case. The window shall be free of flaws which would affect the visibility of the dial and pointer.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:34Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspections shall be classified as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of preparation for delivery (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.3.2 Examination. Samples selected in accordance with 4.3.1 shall be examined for the defects specified in 4.4.1. Acceptable Quality Level (AQL) shall be 2.5 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The multimeter shall be examined as specified herein for the following defects:

101. Case not as specified.
102. Scales illegible or not as specified.
103. Pointer not as specified.
104. Identification marking not as specified.
105. Terminals or leads missing or not as specified.
106. Workmanship not as specified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

4.5 Inspection of preparation for delivery.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

107.	Materials, methods, and containers not as specified for Level A or B. Each incorrect material, method, or container shall be considered one defect.
108.	Quantities intermediate-packaged together not as specified for Level A or B.
109.	Gross weight or size exceeds those specified for Level A or B.
110.	Marking missing, illegible, incorrect, or incomplete for Level A, B, or C.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be Level A, B, or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Unit packaging. Each multimeter shall be preserved in accordance with MIL-P-116, Method IA15. The container shall conform to PPP-B-566 or PPP-B-676.

5.1.1.2 Intermediate packaging. Twenty-five unit packages shall be intermediate packaged together in a close-fitting box conforming to PPP-B-636, W6c or W5c, style optional. Boxes shall be closed in accordance with the appendix to the box specification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:34Z
Check the source to verify that this is the current version before use.

5.1.2 Level B.

5.1.2.1 Unit packaging. Each multimeter shall be preserved in accordance with MIL-P-116, Method IC2. The container shall conform to PPP-B-566 or PPP-B-676.

5.1.2.2 Intermediate packaging. Intermediate packaging shall be as specified in 5.1.1.2 for Level A, except that the fiberboard box shall conform to PPP-B-636, Type CF, Variety SW, Grade 175, style optional.

5.1.3 Level C. The multimeters shall be packaged to provide adequate protection against deterioration and damage from the supplier to the initial destination.

5.2 Packing. Packing shall be Level A, B, or C as specified (see 6.2).

5.2.1 Level A. The multimeters packaged as specified in 5.1 shall be packed together in close-fitting boxes conforming to PPP-B-601, overseas type, Grade B, style optional. The gross weight of each box shall not exceed 200 pounds. The boxes shall be closed and strapped in accordance with the appendix to the box specification.

5.2.2 Level B. The multimeters packaged as specified in 5.1 shall be packed together in close-fitting boxes conforming to PPP-B-636, V3c, style optional; or PPP-B-640, Class 2, style optional. The gross weight or size shall not exceed the limitation of the container. The boxes shall be closed and sealed in accordance with the appendix to the applicable box specification.

5.2.3 Level C. Multimeters preserved and packaged as specified in 5.1 shall be packed to assure carrier acceptance and safe delivery to destination at lowest rating in fiberboard containers complying with Uniform Freight Classification rules or National Motor Freight Classification rules.

5.3 Marking. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:34Z
Check the source to verify that this is the currentversion before use.

6. NOTES

6.1 Intended use. The multimeter is intended for use in measuring current capacities of No. 6 or smaller dry cells, and voltage capacities of dry cells and storage batteries. NSN 6625-00-669-5747 which is used in Equipment Sets 6655-97-CL-E01 and 6675-97-CL-E28 applies thereto. This specification is intended to replace CE PD-130, dated 26 October 1954.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation and packaging and level of packing required (see 5.1 and 5.2). Level B packaging and packing is intended to provide economical but limited protection and should be specified only when it is determined the multimeters will be held in covered storage.

6.3 Information figure. Figure 1 shows a type of multimeter which has been found acceptable; however, the figure is included for illustration only and is not intended to preclude the furnishing of other multimeters which conform to this specification.

Custodian:

Army - ME

Preparing activity:

Army - ME

Project No. 6625-A620

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:34Z
Check the source to verify that this is the current version before use.

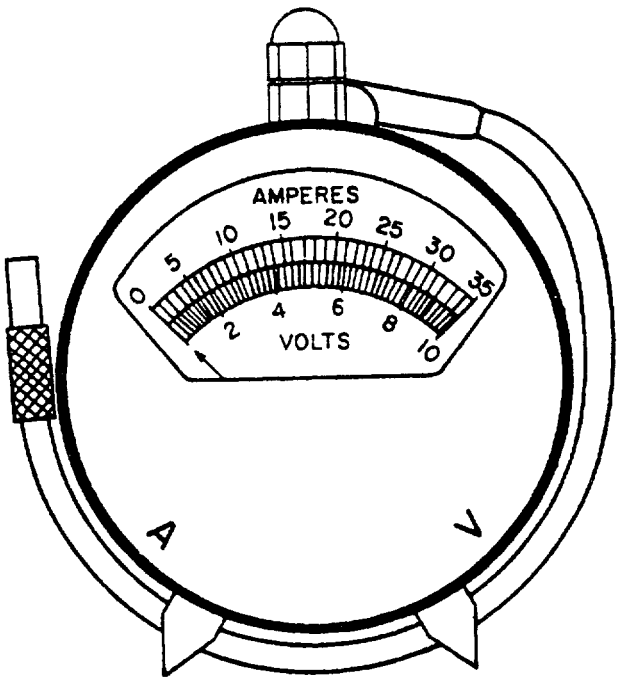


FIGURE 1.- MULTIMETER, BATTERY TESTING.

X-2704

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:34Z
Check the source to verify that this is the current version before use.

TRUCTIONS In a continuing effort to make our standardization documents better, the DoD provides this form for use in mitting comments and suggestions for improvements. All users of military standardization documents are invited to provide -tions This form may be detached, folded along the lines indicated, taped along the loose edge *(DO NOT STAPLE)*, and

In block 5, be as specific as possible about particular problem areas such as wording which required interpretation, was igid, restrictive, loose, ambiguous, or was incompatible, and give proposed wording changes which would alleviate the lems Enter in block 6 any remarks not related to a specific paragraph of the document If block 7 is filled out, an owledgement will be mailed to you within 30 days to let you know that your comments were received and are being idered

E This form may not be used to request copies of documents, nor to request waivers, deviations, or clarification of fication requirements on current contracts Comments submitted on this form do not constitute or imply authorization ive any portion of the referenced document(s) or to amend contractual requirements

(Fold along this line)

(Fold along this line)

RTMENT OF THE ARMY

NO POSTAGE NECESSARY IF MAILED IN THE UNITED STATES

IAL BUSINESS
TY FOR PRIVATE USE $300

BUSINESS REPLY MAIL
FIRST CLASS PERMIT NO 12062 WASHINGTON D C
POSTAGE WILL BE PAID BY THE DEPARTMENT OF THE ARMY

Commander
US Army Mobility Equipment
Research, Development Center
Attn: AMSME-RZD-S
Fort Belvoir, VA 22060

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:34Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
	☐ USER
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ MANUFACTURER
	☐ OTHER *(Specify)*. ____

5. PROBLEM AREAS

a. Paragraph Number and Wording

b. Recommended Wording.

c. Reason/Rationale for Recommendation

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE SUBMITTED *(YYMMDD)*

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:34Z
Check the source to verify that this is the currenversion before use.